

Requirements Gathering

1. Stakeholder Analysis

- **Machine Learning Engineers:** build and maintain the prediction model
- **Maintenance Technicians:** Act on maintenance alerts generated by the system
- **System Administrators:** Manage deployments, data pipelines, and system health
- **End Users:** Input machine sensor readings and view predictions

2. User Stories & Use Cases

- As the **End User**, I want to input readings from sensor so the system can predict if maintenance is needed in the next 7 days
- As the **Maintenance technician**, I need a clear prediction results, so that I can schedule maintenance accordingly
- As the **ML Engineer**, I need to re-train and update the model as that the predictions remain accurate and trustworthy over time
- As the **System Admin**, I need to monitor system uptime so that the service is always available

3. Functional Requirements

- sensor inputs: temperature, vibration, and pressure readings
- Run the trained ML model according to the provided inputs
- Return a binary prediction, maintenance needed or not within 7 days
- Display prediction as binary score (true/false)
- Support model updates

4. Non-Functional Requirements

- Prediction response time under 2 seconds
- System uptime of at least 99%
- Able to handle multiple prediction requests
- Sensor data should be stored and transmitted securely
- Model and codebase should be easy to update
- Prediction output should be simple and understandable for non-technical users